3-DP-Longest Common Subsequence

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer:(penalty regime: 0 %)

Feedback

#include <stdio.h>

#include <string.h>

```
int main() {
   char s1[100], s2[100];

   scanf("%s", s1);
   scanf("%s", s2);

   int m = strlen(s1);
   int n = strlen(s2);

   int dp[101][101] = {0};

   for (int i = 1; i <= m; i++) {
      for (int j = 1; j <= n; j++) {
         if (s1[i - 1] == s2[j - 1]) {
            dp[i][j] = dp[i - 1][j - 1] + 1;
         } else {
            if (dp[i - 1][j] > dp[i][j - 1])
               dp[i][j] = dp[i - 1][j];
            else
               dp[i][j] = dp[i][j - 1];
         }
      }
   }

   printf("%d", dp[m][n]);

   return 0;
}
```

	Input	Expected	Got	
	aab azb	2	2	
	ABCD ABCD	4	4	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.